CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2024 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **What is the need for a Computer Network?**

Need for Computer Networks: Resource sharing (printers, storage servers), centralized data management, cost reduction, electronic communication (email, video conferences), and distributed processing power.

Q2 (3M) **Define HDLC (High-Level Data Link Control).**

Definition: HDLC is a bit-oriented synchronous data link layer protocol developed by ISO that provides reliable point-to-point and multipoint communication using frame delimitation (flag byte `01111110` with bit stuffing), error detection, and sliding window flow control.

Q3 (3M) **Explain the concept of Piggybacking.**

Definition: Piggybacking is a flow control optimization technique where an acknowledgment (ACK) for received data is delayed and attached into the header of the next outgoing data frame sent back to the sender, eliminating the overhead of transmitting independent acknowledgment packets.

Q4 (3M) **Compare Datagram Approach and Virtual Circuit Approach in Packet Switching.**

Comparison:

Aspect	Datagram Approach (Connectionless)	Virtual Circuit Approach (Connection-Oriented)
Call Setup	None; packets transmitted independently.	Setup phase required before transmission.
Path	Each packet routed independently; may arrive out of order.	All packets follow identical pre-configured path in sequence.
Header Overhead	Large (full destination IP on every packet).	Small (short Virtual Circuit Identifier - VCI).

Q5 (3M) **Describe how SMTP protocol is used in E-Mail applications.**

SMTP (Simple Mail Transfer Protocol): A push protocol running over TCP port 25 that transfers email messages from a sender's user agent/client to the sender's mail server, and from the sender's mail server to the recipient's mail server across the Internet.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Discuss the functions of Transport, Presentation, and Application layers in the OSI model.**

Functions:

- **Transport Layer:** End-to-end segmentation, port addressing, connection management, reliable delivery, and flow control.
- **Presentation Layer:** Data translation (syntax formatting), encryption/decryption, and data compression.
- **Application Layer:** Direct user interfaces for network services (HTTP, FTP, SMTP, DNS).

Q7 (7.5M) **Explain the switching methods in Computer Networks.**

Switching Methods: Circuit Switching (dedicated physical communication path established before data transfer), Packet Switching (data split into discrete packets routed dynamically), and Message Switching (entire message stored and forwarded hop-by-hop).

Q8 (7.5M) **Explain in brief: (a) Guided Media, (b) Transmission Modes.**

(a) Guided Media: Physical physical conductors that guide signals along a specific path: Twisted Pair (UTP/STP), Coaxial Cable, and Optical Fiber.
(b) Transmission Modes:

- *Simplex:* Unidirectional transmission (e.g., keyboard to CPU, TV broadcast).
- *Half-Duplex:* Bidirectional, but only one direction at a time (e.g., Walkie-Talkie).
- *Full-Duplex:* Simultaneous bidirectional transmission (e.g., telephone call).

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **What are the layers of the ISO/OSI Protocol Stack? Briefly list their functions.**

Complete OSI Architecture: Physical, Data Link, Network, Transport, Session, Presentation, and Application layers with detailed protocol functions.

Q10 (15M) **Explain in detail about the Error & Flow Control mechanisms employed at the Data Link Layer.**

Mechanisms: Framing methods (byte stuffing, bit stuffing), error detection (CRC-32), and flow control ARQ schemes (Stop-and-Wait, Go-Back-N, Selective Repeat).

Q11 (15M) **(a) Explain Shannon Capacity with examples. (b) What is CSMA? List protocols used with CSMA.**

(a) Shannon Capacity Theorem: Gives theoretical maximum data rate C of a channel with bandwidth B and Signal-to-Noise Ratio SNR:
$C = B \log_2(1 + SNR) \text{ bps}$
Example: For telephone line with B = 3000 Hz and SNR = 3162 (35 dB): $C = 3000 \log_2(1 + 3162) \approx 3000 \times 11.62 = 34,860 \text{ bps} \approx 34.8 \text{ kbps}$.

(b) CSMA Protocols: 1-Persistent CSMA, Non-Persistent CSMA, p-Persistent CSMA, and CSMA/CD / CSMA/CA.

Q12 (15M) **(a) Explain ISDN Architecture and Services. (b) Describe Congestion Control Algorithms.**

(a) Integrated Services Digital Network (ISDN): Transmits voice, video, and data over digital telephone networks. BRI (2B+D = 144 kbps) and PRI (23B+D / 30B+D = 1.544 / 2.048 Mbps).
(b) Congestion Control: Open-loop (prevention via traffic policing) vs Closed-loop (reaction via choke packets and Leaky/Token Bucket algorithms).

Q13 (15M) **Describe any three: (a) SSL, (b) Modems, (c) Line Configuration, (d) Distortion vs Noise.**

Short Notes:

- **SSL (Secure Sockets Layer):** Cryptographic protocol providing end-to-end encryption, authentication, and integrity over TCP.
- **Modems (Modulator-Demodulator):** Converts digital computer signals to analog audio frequencies for transmission over analog lines and vice versa.
- **Line Configuration:** Point-to-Point (dedicated link between two devices) vs Multipoint (multidrop link where multiple devices share a single link).
- **Distortion vs Noise:** Distortion is alteration of signal shape caused by different frequency phase delays in transmission media. Noise is unwanted external thermal/electromagnetic energy added into the signal.
